

An entire function of order five exceeds the planar bound

ziangni-sys

4 October 2026

Conjecture 00000002260. The stated relation $\dim_H J(f) \geq \rho/2$ for entire functions is false without a restriction on the order ρ . Neither source version imposes $\rho \leq 4$.

Take the actual entire function

$$f(z) = \exp(z^5).$$

It is holomorphic on all of $\mathbb{C}$, as the composition of the entire exponential and the polynomial z^5 . We compute its genuine maximum modulus, rather than assume an order value.

For $r \geq 0$ and $|z| \leq r$, the usual exponential norm bound gives

$$|f(z)| = |\exp(z^5)| \leq \exp(|z^5|) = \exp(|z|^5) \leq \exp(r^5).$$

At the positive real point $z = r$, which lies on the circle $|z| = r$, equality holds:

$$|f(r)| = \exp(r^5).$$

Consequently both the closed-disk supremum and the usual circular maximum modulus equal

$$M_f(r) = \exp(r^5).$$

In particular the function's actual entire-function order is

$$\rho(f) = \limsup_{r \rightarrow \infty} \frac{\log \log M_f(r)}{\log r} = 5.$$

Indeed, for every $r > 1$ the displayed quotient is exactly

$$\frac{\log(r^5)}{\log r} = \frac{5 \log r}{\log r} = 5.$$

The complex plane is a real two-dimensional Euclidean space, so its Hausdorff dimension is two. Monotonicity gives the universal statement

$$\dim_H S \leq \dim_H \mathbb{C} = 2 \quad \text{for every subset } S \subseteq \mathbb{C}.$$

Apply this to the genuine Julia set $J(f)$, which is a subset of the complex plane by definition. No computation of its dynamical structure is needed. The proposed lower bound would require

$$\frac{5}{2} = \frac{\rho(f)}{2} \leq \dim_H J(f) \leq 2,$$

which is impossible.

Scope and verification. This disproves the unrestricted order–dimension relation as written. It does not assert a dimension formula for this Julia set, nor refute bounds stated with appropriate restrictions or a different dependence on order. The Lean project proves actual entire differentiability, the maximum-modulus supremum and its exact value, the logarithmic growth limit and actual limsup order, and the Hausdorff bound for *every* subset of $\mathbb{C}$. Its final theorem quantifies over all planar subsets; it does not invent a surrogate definition of a Julia set. Applying this stronger theorem to any genuine $J(f)$ gives the contradiction above. Extended-real Hausdorff dimensions are handled directly in Lean, using $\text{ofReal}(\rho/2)$ and the finite bound two.